Activity 4

Let us take two sachets of juice. In both the sachets, it is written 100ml. Let us empty two sachets and pour the juice into the following glasses.

- Does its shape change? Yes / No

  A liquid needs a container and it takes the shape of a container because the particles slide over one another and keep moving.

- Does its volume change when it is poured into a big glass as well as a small one? Yes / No

  The amount of juice is the same in both glasses.

- How will you find out whether the volume has changed or not?

  The volume of a liquid remains the same whether it is kept in a large container or a small one but its shape changes.

Activity 5

Lift an uninflated cycle tube. Inflate it and now lift it again. Is there a change in the weight? Can we say that air has mass?

We can say that air is also a matter. Though we cannot see it, it occupies space and also has mass. Let us try to know more about matter.

Test Yourself

1 Name an object which is brittle and transparent. _____

2. Name an object which can be stretched. _____

3. Name two objects which can be bent. _____

3.4 Compressibility of gases compared to liquids and solids

Let us take three identical syringes. Close the nozzles tightly with a cork. After removing the plunger first let us fill it with fine chalk powder. Try to press plunger down. What do you observe?

Now let us fill the second one with water. Press the plunger down. What do you observe? Let us now draw the piston back to suck air into the third one. Press the plunger down. What do you observe? Is it easy or hard to press? Record your observations and share among the group members.

You would have observed that the plunger moved freely in syringe with air than in water. It was difficult to press the liquids and the piston hardly moved in chalk powder. Thus, we can conclude that gases are highly compressible as compared to liquids and solids.